

5G RAN

3D Networking Experience Improvement Feature Parameter Description

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For vulnerability information, enterprise customers can visit the following web page:

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Contents

1 Change History.....1

1.1 5G RAN10.1 Draft A (2025-12-31).....1

2 About This Document.....2

2.1 General Statements.....2

2.2 Features in This Document.....2

2.3 Differences Between NSA and SA.....3

2.4 Differences Between High Frequency Bands and Low Frequency Bands.....3

3 Overview.....4

4 Intra-Frequency 3D Networking Experience Improvement.....6

4.1 Intra-Frequency Common Channel Interference Avoidance.....6

4.1.1 Principles.....6

4.1.2 Network Analysis.....9

4.1.2.1 Benefits.....9

4.1.2.2 Impacts.....10

4.1.3 Requirements.....11

4.1.3.1 Licenses.....11

4.1.3.2 Functions.....12

4.1.3.2.1 Prerequisite Functions.....12

4.1.3.2.2 Mutually Exclusive Functions.....13

4.1.3.2.3 Function Impacts.....14

4.1.3.3 Hardware.....16

4.1.3.4 Cells.....17

4.1.3.5 Others.....17

4.1.4 Operation and Maintenance.....17

4.1.4.1 Precautions.....17

4.1.4.2 Data Configuration.....17

4.1.4.2.1 Data Preparation.....17

4.1.4.2.2 Using MML Commands.....18

4.1.4.2.3 Using the MAE-Deployment.....19

4.1.4.3 Activation Verification.....19

4.1.4.4 Network Monitoring.....20

4.2 Interference Avoidance for Indoor Cells.....20

4.2.1 Principles..... 20

4.2.2 Network Analysis..... 22

4.2.2.1 Benefits..... 22

4.2.2.2 Impacts..... 23

4.2.3 Requirements..... 23

4.2.3.1 Licenses..... 23

4.2.3.2 Functions..... 23

4.2.3.2.1 Prerequisite Functions..... 23

4.2.3.2.2 Mutually Exclusive Functions..... 24

4.2.3.2.3 Function Impacts..... 25

4.2.3.3 Hardware..... 26

4.2.3.4 Others..... 26

4.2.4 Operation and Maintenance..... 27

4.2.4.1 Data Configuration..... 27

4.2.4.1.1 Data Preparation..... 27

4.2.4.1.2 Using MML Commands..... 27

4.2.4.1.3 Using the MAE-Deployment..... 27

4.2.4.2 Activation Verification..... 28

4.2.4.3 Network Monitoring..... 28

4.3 Enhanced Interference Avoidance for Indoor Cells..... 28

4.3.1 Principles..... 28

4.3.2 Network Analysis..... 30

4.3.2.1 Benefits..... 30

4.3.2.2 Impacts..... 31

4.3.3 Requirements..... 31

4.3.3.1 Licenses..... 31

4.3.3.2 Functions..... 32

4.3.3.2.1 Prerequisite Functions..... 32

4.3.3.2.2 Mutually Exclusive Functions..... 32

4.3.3.2.3 Function Impacts..... 34

4.3.3.3 Hardware..... 35

4.3.3.4 Cells..... 35

4.3.3.5 Others..... 36

4.3.4 Operation and Maintenance..... 36

4.3.4.1 Data Configuration..... 36

4.3.4.1.1 Data Preparation..... 36

4.3.4.1.2 Using MML Commands..... 36

4.3.4.1.3 Using the MAE-Deployment..... 37

4.3.4.2 Activation Verification..... 37

4.3.4.3 Network Monitoring..... 37

5 Inter-Frequency 3D Networking Experience Improvement.....39

5.1 Principles..... 39

5.2 Network Analysis..... 43

5.2.1 Benefits..... 43

5.2.2 Impacts..... 44

5.3 Requirements..... 45

5.3.1 Licenses..... 45

5.3.2 Functions..... 46

5.3.2.1 Prerequisite Functions..... 46

5.3.2.2 Mutually Exclusive Functions..... 48

5.3.2.3 Function Impacts..... 49

5.3.3 Hardware..... 51

5.3.4 Cells..... 52

5.3.5 Others..... 52

5.4 Operation and Maintenance..... 52

5.4.1 Data Configuration..... 52

5.4.1.1 Data Preparation..... 52

5.4.1.2 Using MML Commands..... 53

5.4.1.3 Using the MAE-Deployment..... 54

5.4.2 Activation Verification..... 54

5.4.3 Network Monitoring..... 54

6 Parameters..... 55

7 Counters..... 57

8 Glossary..... 58

9 Reference Documents..... 59

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-11-07).

Technical Changes

None

Editorial Changes

Added an illustration of 3D networking. For details, see [Figure 3-1](#).

Changed "intra-frequency 3D networking experience improvement" to "intra-frequency common channel interference avoidance" in certain descriptions. For details, see [4.1 Intra-Frequency Common Channel Interference Avoidance](#).

Added sections [4.1.3.4 Cells](#) and [5.3.4 Cells](#).

Removed the "Parameters used for optimization" table from [4.1.4.2.1 Data Preparation](#).

Added section [4.2.3.4 Others](#).

Revised descriptions of the background of enhanced interference avoidance for indoor cells. For details, see [4.3.1 Principles](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-060202	3D Networking Experience Improvement	4 Intra-Frequency 3D Networking Experience Improvement 5 Inter-Frequency 3D Networking Experience Improvement

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Intra-frequency 3D networking experience improvement	None	4 Intra-Frequency 3D Networking Experience Improvement
Inter-frequency 3D networking experience improvement	None	5 Inter-Frequency 3D Networking Experience Improvement

2.4 Differences Between High Frequency Bands and Low Frequency Bands

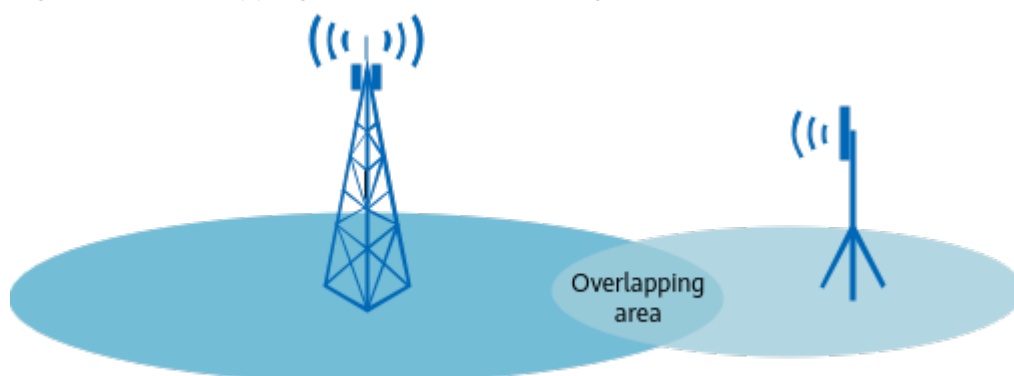
This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Intra-frequency 3D networking experience improvement	Supported only in low frequency bands	4 Intra-Frequency 3D Networking Experience Improvement
Inter-frequency 3D networking experience improvement	Supported only in low frequency bands	5 Inter-Frequency 3D Networking Experience Improvement

3 Overview

In indoor-outdoor or macro-pole continuous networking (3D networking), coverage overlaps between cells ([Figure 3-1](#)). In these overlapping areas, interference can occur in two situations: (1) If the serving cell's PDSCH bandwidth overlaps with a neighboring cell's common channel bandwidth and the two cells use different common channel configurations, the serving cell's PDSCH experiences interference from the neighboring cell's common channel (SSB, RMSI, and cell-level periodic CSI-RS); (2) If adjacent cells have overlapping time-frequency positions for downlink scheduling, their PDSCHs interfere with each other. To address this issue, the 3D Networking Experience Improvement feature is introduced.

Figure 3-1 Overlapping area in 3D networking



The 3D Networking Experience Improvement feature is classified into intra-frequency and inter-frequency 3D networking experience improvement, depending on whether adjacent cells use the same SSB frequency. [Table 3-1](#) summarizes the functions included in this feature.

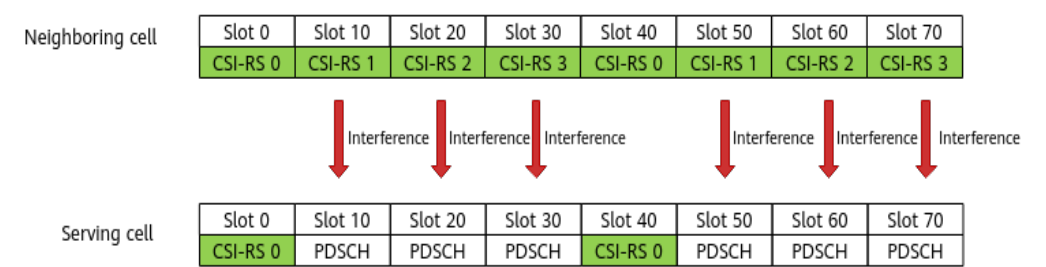
Table 3-1 Functions of the 3D Networking Experience Improvement feature

Type	Function	Interference Source	Reference
Intra-frequency 3D networking experience improvement	Enhanced intra-frequency SSB interference avoidance	SSB	4.1 Intra-Frequency Common Channel Interference Avoidance
	Intra-frequency CSI-RS interference avoidance	CSI-RS	
	Interference avoidance for indoor cells	PDSCH	4.2 Interference Avoidance for Indoor Cells
	Enhanced interference avoidance for indoor cells	PDSCH	4.3 Enhanced Interference Avoidance for Indoor Cells
Inter-frequency 3D networking experience improvement	Inter-frequency SSB interference avoidance	SSB	5 Inter-Frequency 3D Networking Experience Improvement
	Enhanced inter-frequency SSB interference avoidance	SSB	
	Inter-frequency CSI-RS interference avoidance	CSI-RS	

 NOTE

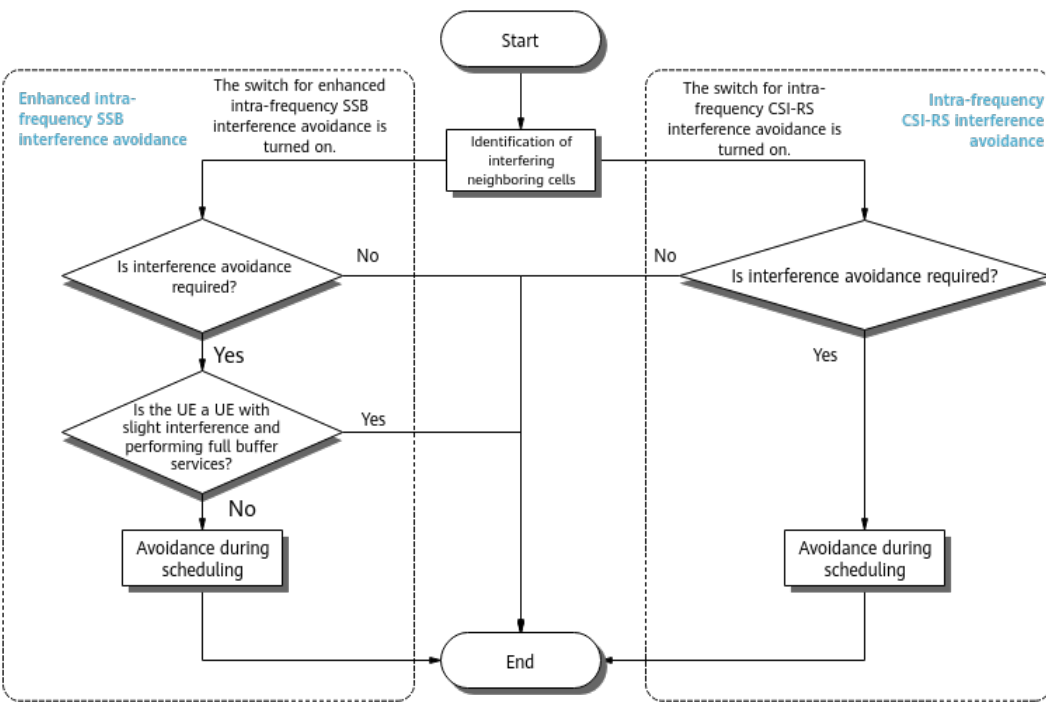
- An overlapping area denotes the overlap between the coverage of the serving cell and that of a neighboring cell.
- In this document, the serving cell is defined as a cell that experiences interference from a neighboring cell's common channel or PDSCH and therefore requires the 3D Networking Experience Improvement feature.

Figure 4-2 CSI-RS interference from a neighboring cell (example: identical 40-slot CSI-RS period)



Based on the types of interfering signals, intra-frequency common channel interference avoidance is classified into intra-frequency CSI-RS interference avoidance and enhanced intra-frequency SSB interference avoidance. [Figure 4-3](#) shows how intra-frequency common channel interference avoidance works.

Figure 4-3 Process of intra-frequency common channel interference avoidance



Identification of Interfering Neighboring Cells

The **INTRA_FREQ_NCELL_STAT_SW** option of the **NRDUCellCollabServ.NeighborNrCellMgmtSw** parameter specifies whether to enable intra-frequency neighboring cell measurement. After this option is selected, UEs report A3 measurements of intra-frequency neighboring cells based on the A3 threshold (specified by **NRCCellComp.IntraFreqNCellStatA3Offset**). The serving cell then ranks these neighboring cells by the measurements and selects the top five interfering neighboring cells. Note that this option needs to be selected for both the serving and neighboring cells. Any neighboring cell with this option deselected is excluded from this ranking and subsequent procedures.

 NOTE

- Interfering neighboring cells are identified every 30 minutes.
- After intra-frequency CSI-RS interference avoidance or enhanced intra-frequency SSB interference avoidance is enabled, the system first identifies interfering neighboring cells and then initiates interference avoidance once such cells are detected.

Enhanced Intra-Frequency SSB Interference Avoidance

This function is controlled by the **INTRA_FREQ_SSB_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch** parameter. After this option is selected, this function works as follows:

1. Determining whether interference avoidance is required
 - If enhanced intra-frequency SSB interference avoidance is enabled for a neighboring cell, the serving cell obtains SSB configurations of the neighboring cell over the Xn interface. Based on this information, the serving cell determines whether its PDSCH experiences SSB interference from the neighboring cell as follows: If SSB time-domain positions are inconsistent between the two cells, the serving cell's PDSCH experiences SSB interference from the neighboring cell, in which case interference avoidance is required. If SSB time-domain positions are consistent between the two cells, the serving cell's PDSCH does not experience SSB interference from the neighboring cell, in which case interference avoidance is not required and normal scheduling is applied.
 - If enhanced intra-frequency SSB interference avoidance is disabled for a neighboring cell, the serving cell cannot obtain SSB configurations of the neighboring cell. As a result, interference from the neighboring cell cannot be avoided.
2. Checking whether a UE is a full-buffer UE under slight-interference conditions
To check whether a full-buffer UE is under slight-interference conditions, compare the UE's serving-cell and interfering-neighboring-cell signal strengths. If the signal strength difference exceeds the specified threshold, the UE is considered as experiencing slight interference from the neighboring cell. Otherwise, the UE is considered as experiencing strong interference from the neighboring cell.
3. Avoidance during scheduling
For all UEs except full-buffer UEs under slight-interference conditions, PDSCH scheduling avoids the time-frequency positions used by neighboring cells for SSB transmission whenever possible.

Intra-Frequency CSI-RS Interference Avoidance

This function is controlled by the **INTRA_FREQ_CSIRS_AVOID_SW** option of the **NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch** parameter. After this option is selected, this function works as follows:

1. Determining whether interference avoidance is required
 - If intra-frequency CSI-RS interference avoidance is enabled for a neighboring cell, the serving cell obtains CSI-RS configurations of the neighboring cell over the Xn interface. Based on this information, the serving cell determines whether its PDSCH experiences CSI-RS

interference from the neighboring cell as follows: If CSI-RS time-domain positions are inconsistent between the two cells, the serving cell's PDSCH experiences CSI-RS interference from the neighboring cell, in which case interference avoidance is required. If CSI-RS time-domain positions are consistent between the two cells, the serving cell's PDSCH does not experience CSI-RS interference from the neighboring cell, in which case interference avoidance is not required and normal scheduling is applied.

- If intra-frequency CSI-RS interference avoidance is disabled for a neighboring cell, the serving cell cannot obtain CSI-RS configurations of the neighboring cell. As a result, interference from the neighboring cell cannot be avoided.
2. Avoidance during scheduling
- For all UEs, PDSCH scheduling avoids the time-frequency positions used by neighboring cells for CSI-RS transmission whenever possible. This function helps avoid interference caused only by cell-level periodic CSI-RS.

4.1.2 Network Analysis

4.1.2.1 Benefits

Enhanced Intra-Frequency SSB Interference Avoidance

The analysis of benefits in NR TDD cells is as follows:

In drive tests (DTs) in intra-frequency 3D continuous networking scenarios, you are advised to enable enhanced intra-frequency SSB interference avoidance if all of the following conditions are met: Cells are configured with different quantities of SSB beams; the downlink block error rate (BLER) of the serving cell increases significantly (by 20%) in the slots with transmission of the neighboring cells' SSBs. In this case, CSI-RS interference also exists. It is recommended that intra-frequency CSI-RS interference avoidance be also enabled. The SSB beam configurations of different cells are determined by querying the configured values of related parameters in section "Basic Beam Management" in *Beam Management*.

- When full-buffer services are being performed:
 - For stationary UEs whose PDSCHs suffer SSB interference from neighboring cells, the average downlink throughput increases by more than 10%.
 - For UEs in DTs, the increase in the average downlink throughput is related to the proportion of geographic locations where the UEs suffer SSB interference from neighboring cells in the DT routes. The larger the proportion, the higher the increase in the average downlink throughput of UEs in DTs.
- When burst services are performed, the gains of this function are related to the time-frequency domain positions for scheduling. The specific gains cannot be determined, but there is no negative impact.

Intra-Frequency CSI-RS Interference Avoidance

In intra-frequency 3D continuous networking scenarios, you are advised to enable intra-frequency CSI-RS interference avoidance to reduce the interference from

neighboring cells and increase the downlink throughput of UEs in the overlapping areas if all of the following conditions are met: The time-domain position configurations of cell-level periodic CSI-RSs are inconsistent between cells; the downlink BLERs of the serving cell in the slots with transmission of the neighboring cells' CSI-RSs increase significantly (by 20%). In this case, SSB interference also exists. It is recommended that enhanced intra-frequency SSB interference avoidance be also enabled.

After intra-frequency CSI-RS interference avoidance is enabled, the average downlink throughput increases by up to 8% for beneficiary UEs in the overlapping areas between TDD cells. The gains are related to such factors as the traffic model and network load. The larger the data packet size, the greater the gains. The higher the network load, the greater the gains. This function is recommended when the serving cell is in medium/heavy load (downlink PRB usage > 30%). The time domain positions of cell-level periodic CSI-RSs are determined by **NRDUCellCsirs.CsiPeriod** and **NRDUCellCsirs.CsirsCellResourceNum**.

Besides intra-frequency CSI-RS interference avoidance, static avoidance of CSI-RS interference and inter-cell CSI-RS for CM interference avoidance (see *Channel Management*) also help avoid CSI-RS interference. However, the CSI-RS configuration information of neighboring cells is not obtained in the processing of the latter two functions, whereby interference avoidance is performed based on the maximum available time-frequency resources during scheduling, leading to relatively high time-frequency resource consumption. Therefore, you are advised to enable only intra-frequency CSI-RS interference avoidance when there is CSI-RS interference from neighboring cells.

NOTE

- Static avoidance of CSI-RS interference is controlled by the **CSIRS_INTRF_STATIC_AVOID_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter.
- Inter-cell CSI-RS for CM interference avoidance is controlled by the **NRDUCELLCSIRS.NCellCsirsIntrfAvoid** parameter.
- For details about related indicators, see [4.1.4.4 Network Monitoring](#).

4.1.2.2 Impacts

The impacts between this function and other functions are described in [4.1.3.2.3 Function Impacts](#).

Enhanced Intra-Frequency SSB Interference Avoidance

- A cell will automatically restart after the setting of the option for this function is changed. If this function is enabled for an indoor distributed base station or pole-mounted base station, it is recommended that this function also be enabled for adjacent macro base stations. Doing so lessens the impact of cell restarts on the network.
- This function reduces the time-frequency resources available for downlink scheduling. The degree of the reduction depends on these configurations: cell bandwidth, and neighboring cell's SSB resource quantity and SSB period. For a TDD cell with a bandwidth of 100 MHz, available RBs are reduced by up to 7%. As such, the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) decreases by up to 7% for full buffer services.

Intra-Frequency CSI-RS Interference Avoidance

- A cell will automatically restart after the setting of the option for this function is changed. If this function is enabled for an indoor distributed base station or pole-mounted base station, it is recommended that this function also be enabled for adjacent macro base stations. Doing so lessens the impact of cell restarts on the network.
- This function reduces the time-frequency resources available for downlink scheduling. The degree of the reduction depends on these configurations: numbers of CSI-RS ports for the serving cell and for neighboring cells, and difference in cell-level CSI-RS time-frequency resources between the serving and neighboring cells. Available RBs are reduced by up to 2%. As such, the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) decreases by up to 2% for full buffer services.

4.1.3 Requirements

4.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-060202	3D Networking Experience Improvement	NR0S03DNEM00	Per Cell
For the license control item NR0S03DNEM00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the INTRA_FREQ_SSB_AVOID_ENH_SW or INTRA_FREQ_CSIRS_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter cannot be modified due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.			

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

Enhanced Intra-Frequency SSB Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	Intra-frequency neighboring cell measurement	INTRA_FREQ_ NCELL_STAT_S W option of the NRDUCellColla bServ.Neighbo rNrCellMgmtS w parameter	None	The switch for intra- frequency neighboring cell measurement must be turned on.

Intra-Frequency CSI-RS Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	Intra-frequency neighboring cell measurement	INTRA_FREQ_ NCELL_STAT_S W option of the NRDUCellColla bServ.Neighbo rNrCellMgmtS w parameter	None	The switch for intra- frequency neighboring cell measurement must be turned on.
Lo w- freq uen cy TD D	CSI-RS rate matching	CSIRS_RATEM ATCH_SW option of the NRDUCellPdsc h.RateMatchS witch parameter	<i>Downlink Scheduling</i>	CSI-RS rate matching must be enabled.

4.1.3.2.2 Mutually Exclusive Functions

Enhanced Intra-Frequency SSB Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	DL Blanking	NRDUCell.Peer CellSlotAssign ment	<i>Standards Complianc e</i>	The NRDUCell.PeerCellSlot Assignment parameter must be set to NOT_CONFIG .
Lo w- freq uen cy TD D	Inter-TRP SSB staggering	INTER_TRP_SS B_STAGGER_S WITCH option of the NRDUCellMult iTrp.MultiTrpA lgoSwitch parameter	<i>Hyper Cell Cell Combinati on</i>	These two functions are mutually exclusive: inter-TRP SSB staggering and enhanced intra-frequency SSB interference avoidance.

Intra-Frequency CSI-RS Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	DL Blanking	NRDUCell.Peer CellSlotAssign ment	<i>Standards Complianc e</i>	The NRDUCell.PeerCellSlot Assignment parameter must be set to NOT_CONFIG .
Lo w- freq uen cy TD D	Time-domain reservation mode	NRDUCellTime Intrf.TimeDom ainRsvMode	None	Intra-frequency CSI-RS interference avoidance cannot be enabled when the NRDUCellTimeIntrf.Ti meDomainRsvMode parameter is set to TIME_DOMAIN_RESERV ED_MODE_1 , TIME_DOMAIN_RESERV ED_MODE_2 , or TIME_DOMAIN_RESERV ED_MODE_3 .

4.1.3.2.3 Function Impacts

Enhanced Intra-Frequency SSB Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- fre que ncy TD D	Downlink intra- band CA	INTRA_BAND_CA_SW option of the NRDUCellAI goSwitch.C aAlgoSwitch parameter	<i>Carrier Aggregation</i>	When downlink intra- band CA is enabled, enhanced intra- frequency SSB interference avoidance does not take effect for CA UEs in their secondary cells (SCells).
Lo w- fre que ncy TD D	Downlink intra- FR inter-band CA	INTRA_FR_I NTER_BAND_CA_SW option of the NRDUCellAI goSwitch.C aAlgoSwitch parameter	<i>Carrier Aggregation</i>	When downlink intra-FR inter-band CA is enabled, enhanced intra-frequency SSB interference avoidance does not take effect for CA UEs in their SCells.
Lo w- fre que ncy TD D	Inter-gNodeB CA	INTER_GNO DEB_CA_SW option of the NRDUCellAI goSwitch.C aAlgoSwitch parameter	<i>Carrier Aggregation</i>	When inter-gNodeB CA is enabled, enhanced intra-frequency SSB interference avoidance does not take effect for CA UEs in their SCells.
Lo w- fre que ncy TD D	Common channel interference avoidance	SSB_INTRF_ AVOID_SW option of the NRDUCellAI goSwitch.C ommChnInt rfAvoidSwit ch parameter	<i>Interference Avoidance</i>	Only enhanced intra- frequency SSB interference avoidance takes effect if this function and common channel interference avoidance are both enabled.
Lo w- fre que ncy TD D	Cell Combination	NRDUCell.N rDuCellNet workingMo de set to HYPER_CEL L_COMBINE _MODE	<i>Cell Combination</i>	In a combined cell, interference avoidance during scheduling can work for full buffer services running on UEs that are slightly affected by interference from neighboring cells.

RA T	Function Name	Function Switch	Reference	Description
Lo w- fre que ncy TD D	High-speed Railway Superior Experience	NRDUCell.H ighSpeedFl ag set to HIGH_SPEE D	<i>High Speed Mobility</i>	When High-speed Railway Superior Experience takes effect, interference avoidance during scheduling can work for full buffer services running on UEs that are slightly affected by interference from neighboring cells.
Lo w- fre que ncy TD D	Downlink time- domain MCS adjustment	DL_TD_MCS _ADJ_SW option of the NRDUCellP dsch.DLink AdaptAlgoS witch parameter	None	For indoor distributed base stations or pole- mounted base stations, the benefits delivered by enhanced intra- frequency SSB interference avoidance decrease after downlink time-domain MCS adjustment is enabled.
Lo w- fre que ncy TD D	Intelligent scheduling for power saving	POWER_SA VING_SCHE DULE_SW option of the NRDUCellAl goSwitch.P owerSaving Switch parameter	<i>Energy Conservation and Emission Reduction</i>	If intelligent scheduling for power saving is enabled for a hyper cell and the RMSI intervals are different between TRPs, the serving cell's PDSCH gives way to the RMSI of all TRPs in the hyper cell.

Intra-Frequency CSI-RS Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	Downlink intra- band CA	INTRA_BAND_ CA_SW option of the NRDUCellAlgo Switch.CaAlgo Switch parameter	<i>Carrier Aggregatio n</i>	When downlink intra- band CA is enabled, intra-frequency CSI-RS interference avoidance does not take effect for CA UEs in their SCells.

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTE R_BAND_CA_ SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When downlink intra-FR inter-band CA is enabled, intra-frequency CSI-RS interference avoidance does not take effect for CA UEs in their SCells.
Lo w- freq uen cy TDD	Inter-gNodeB CA	INTER_GNODE B_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When inter-gNodeB CA is enabled, intra-frequency CSI-RS interference avoidance does not take effect for CA UEs in their SCells.
Lo w- freq uen cy TDD	Downlink time-domain MCS adjustment	DL_TD_MCS_A DJ_SW option of the NRDUCellPdsc h.DLLinkAdaptAlgoSwitch parameter	None	For indoor distributed base stations or pole-mounted base stations, the benefits delivered by intra-frequency CSI-RS interference avoidance decrease after downlink time-domain MCS adjustment is enabled.

4.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

For NR TDD:

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.1.3.4 Cells

NR TDD: The serving cell and interfering neighboring cells must have the same bandwidth, slot configuration, CP, and SCS.

Intra-frequency CSI-RS interference avoidance requires that the **NRDUCell.SlotAssignment** parameter not be set to **3_7_DSUUUUSUUU**.

4.1.3.5 Others

The Xn and eXn interfaces of the gNodeB are normal.

4.1.4 Operation and Maintenance

4.1.4.1 Precautions

Indicators related to the throughput of UEs in overlapping areas take effect only after the **INTRA_GNB_OVERLAPPING_MEAS_SW** and **INTER_GNB_OVERLAPPING_MEAS_SW** options of the **NRDUCellAlgoSwitch.CompSwitch** parameter are selected. In addition, the **NRCellComp.IntraFreqNCellStatA3Offset** parameter must be configured. It is good practice to configure the preceding parameters one week before intra-frequency 3D networking experience improvement is enabled. This ensures that sufficient statistics are collected for throughput comparison before and after this function is enabled.

4.1.4.2 Data Configuration

4.1.4.2.1 Data Preparation

Table 4-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Neighboring NR Cell Management Switch	NRDUCellCollab Serv.NeighborNr CellMgmtSw	INTRA_FREQ_NCELL_STAT_SW	Select this option for both the serving and neighboring cells.

Parameter Name	Parameter ID	Option	Setting Notes
Common Channel Interference Avoid Switch	NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch	INTRA_FREQ_SSB_AVOID_ENH_SW	Select this option for both the serving and neighboring cells.
Common Channel Interference Avoid Switch	NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch	INTRA_FREQ_CSIRS_AVOID_SW	Select this option for both the serving and neighboring cells.
Intra-Freq Neighbor Cell Statistics A3 Offset	NRCellComp.IntraFreqNCellStatA3Offset	None	Retain the default value.

4.1.4.2.2 Using MML Commands

Before using MML commands, refer to [4.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling intra-frequency neighboring cell measurement
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, NeighborNrCellMgmtSw = INTRA_FREQ_NCELL_STAT_SW-1;

//Enabling enhanced intra-frequency SSB interference avoidance (Changing the setting of the
INTRA_FREQ_SSB_AVOID_ENH_SW option will cause the cell to automatically restart.)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CommChnIntrfAvoidSwitch=
INTRA_FREQ_SSB_AVOID_ENH_SW-1;

//Enabling intra-frequency CSI-RS interference avoidance (Changing the setting of the
INTRA_FREQ_CSIRS_AVOID_SW option will cause the cell to automatically restart.)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CommChnIntrfAvoidSwitch=
INTRA_FREQ_CSIRS_AVOID_SW-1;
//Configuring the IntraFreqNCellStatA3Offset parameter
MOD NRCELLCOMP: NrCellId=0, IntraFreqNCellStatA3Offset=-30;
```

NOTE

In the preceding three functions, the relevant options need to be selected for both the serving and neighboring cells.

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling enhanced intra-frequency SSB interference avoidance (Changing the setting of the  
INTRA_FREQ_SSB_AVOID_ENH_SW option will cause the cell to automatically restart.)  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
CommChnIntrfAvoidSwitch=INTRA_FREQ_SSB_AVOID_ENH_SW-0;  
  
//Disabling intra-frequency CSI-RS interference avoidance (Changing the setting of the  
INTRA_FREQ_CSIRS_AVOID_SW option will cause the cell to automatically restart.)  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
CommChnIntrfAvoidSwitch=INTRA_FREQ_CSIRS_AVOID_SW-0;  
  
//Disabling intra-frequency neighboring cell measurement  
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, NeighborNrCellMgmtSw=INTRA_FREQ_NCELL_STAT_SW-0;
```

4.1.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.3 Activation Verification

Enhanced Intra-Frequency SSB Interference Avoidance

When the number of SSB beams differs between the serving cell and a neighboring cell, the following activation verification logic applies: If scheduling in the serving cell does not occur in the time-frequency positions where the neighboring cell sends SSB but the serving cell does not, the number of RBs scheduled for full-buffer UEs decreases, indicating that this function has taken effect. A DT tool can verify this decrease. For details about this decrease, see [4.1.2.2 Impacts](#).

NOTE

The number of SSB beams can be obtained from the **ssb-PositionsInBurst** field in the **ServingCellConfigCommonSIB** message. For details, see section 6.3.2 "Radio resource control information elements" in 3GPP TS 38.331 V16.0.0.

Intra-Frequency CSI-RS Interference Avoidance

Start Uu signaling tracing as follows: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**. Locate the **RRCReconfiguration** message and check the values of specific fields in **ZP-CSI-RS-Resource > CSI-RS-ResourceMapping** (see the area enclosed in a red box in [Figure 4-4](#)). If any value changes, this function has taken effect.

Figure 4-4 CSI-RS-ResourceMapping IE

```

CSI-RS-ResourceMapping ::= SEQUENCE {
    frequencyDomainAllocation CHOICE {
        row1 BIT STRING (SIZE (4)),
        row2 BIT STRING (SIZE (12)),
        row4 BIT STRING (SIZE (3)),
        other BIT STRING (SIZE (6))
    },
    nrofPorts ENUMERATED {p1,p2,p4,p8,p12,p16,p24,p32},
    firstOFDMsymbolInTimeDomain INTEGER (0..13),
    firstOFDMsymbolInTimeDomain2 INTEGER (2..12),
    cdn-Type ENUMERATED {noCDM, fd-CDM2, cdn4-FD2-ID2, cdn8-FD2-ID4},
    density CHOICE {
        dot5 ENUMERATED {evenPRBs, oddPRBs},
        one NULL,
        three NULL,
        spare NULL
    },
    freqBand CSI-FrequencyOccupation
}

```

4.1.4.4 Network Monitoring

Intra-frequency common channel interference avoidance increases downlink throughput of UEs in overlapping areas. To verify the benefits of this function, compare the result derived from the counters listed below before and after this function is enabled. Before that, set the necessary parameters following the instructions in [4.1.4.1 Precautions](#), as the measurement of the counters depends on those settings.

Downlink throughput of UEs in overlapping areas = $(N.ThpVol.DL.CellOverlap - N.ThpVol.DL.LastSlot.CellOverlap) / N.ThpTime.DL.RmvLastSlot.CellOverlap$

NOTE

The **N.User.RRCCConn.Avg.CellOverlap** counter can be used to monitor the average number of UEs in RRC_CONNECTED mode in the overlapping area in a common cell.

4.2 Interference Avoidance for Indoor Cells

4.2.1 Principles

Interference occurs between an indoor cell and an outdoor cell when their RB positions for downlink scheduling overlap ([Figure 4-5](#)). When both cells share the same PCI mod 3 value, their scheduling positions may still overlap even after downlink interference randomization-based scheduling (described in *Downlink Scheduling*) is enabled ([Figure 4-6](#)). If this is the case, interference still persists.

Figure 4-5 Interference in an indoor cell

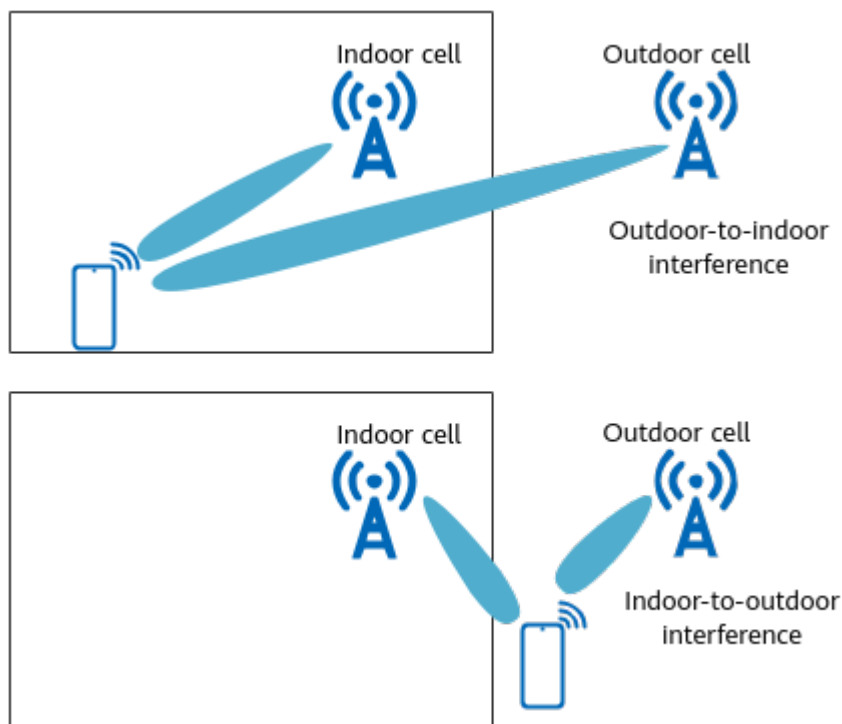
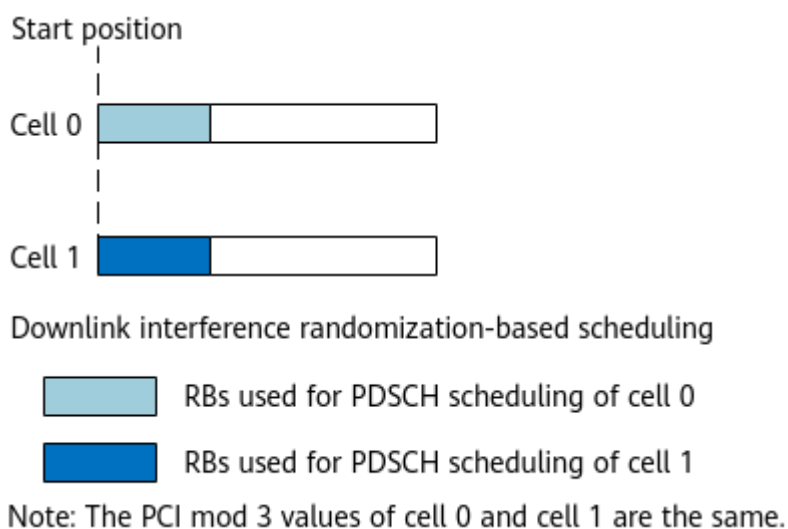


Figure 4-6 Downlink interference randomization-based scheduling



To address the aforementioned issue, interference avoidance for indoor cells is introduced. With this function, an indoor cell proactively avoids using the same frequency-domain positions as those used by outdoor cells for PDSCH transmission to mitigate interference, as shown in [Figure 4-7](#).

This function is controlled by the **INDOOR_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch.IndoorOutdoorCooperateSw** parameter. After this function is enabled:

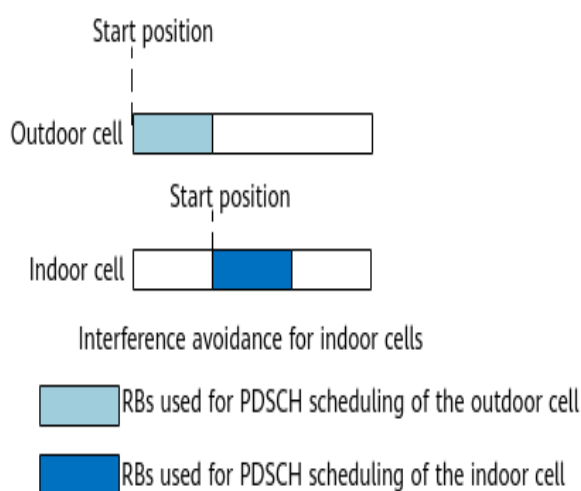
- The indoor cell proactively collects information about strongly interfering neighboring cells based on UE-reported A3 measurements on an hourly basis.

(At least 10 A3 measurements must be reported within one hour for this collection to occur; otherwise, no neighboring cell information is collected for that hour.)

- The indoor cell obtains the PDSCH start-RB allocations of the top three interfering neighboring cells over the Xn interface. Based on this information, it schedules its PDSCH to start at a different RB position, shifting its PDSCH away from the PDSCH in those neighboring cells whenever possible.

In this function, indoor cells are defined as cells served by LampSite base stations or macro base stations that provide distributed antenna system (DAS)-based indoor coverage. (For details, see *Indoor Coverage*.) Note that indoor cells and interfering neighboring cells cannot be deployed on the same base station.

Figure 4-7 Interference avoidance for indoor cells



4.2.2 Network Analysis

4.2.2.1 Benefits

Deployment Suggestions

It is recommended that this function be enabled in isolated indoor cells—each serving a single floor and typically experiencing interference from outdoor cells. If multiple indoor cells are deployed for contiguous coverage (with all neighboring cells being indoor cells), it is not recommended that this function be enabled, because interference between these cells predominates and cannot be effectively mitigated through this function. Additionally, it is not recommended that this function be enabled in outdoor cells.

Benefits

- This function mitigates interference between indoor and outdoor cells under light-load conditions with burst services if (1) downlink interference randomization-based scheduling is disabled or (2) downlink interference randomization-based scheduling is enabled and these cells have the same PCI mod 3 value. This function improves the indoor cell's downlink spectral

efficiency and increases the average downlink throughput of UEs in the overlapping areas between these cells by 2%–3%.

- Under very light load, time-frequency scheduling positions are staggered across cells, in which case interference is low and this function may offer no benefits.

NOTE

- Typically, a cell is considered lightly loaded if its downlink PRB usage is less than or equal to 30%. The downlink PRB usage of a cell is calculated as follows:
 $(N.PRB.DL.Used.Avg / N.PRB.DL.Avail.Avg) \times 100\%$.
- The downlink spectral efficiency of a cell is calculated as follows: $N.ThpVol.DL / (N.PRB.DL.DrbUsed.Avg \times N.DL.PDSCH.Tti.Num)$, expressed as bit/(TTI·PRB).
To express the downlink spectral efficiency of a cell as bits per second per Hz, the preceding formula can be changed as follows: $N.ThpVol.DL / (N.PRB.DL.DrbUsed.Avg \times N.DL.PDSCH.Tti.Num \times 2000 / 360000)$. The idea behind this formula is as follows: 2000 is introduced for a TTI-to-second conversion, because one second has 2000 TTIs (1s = 1000 ms, 1 TTI = 0.5 ms). 360000 is introduced for a PRB-to-Hz conversion, because one PRB spans 360,000 Hz in the frequency domain (1 PRB = 12 subcarriers, 1 subcarrier = 30,000 Hz).

4.2.2.2 Impacts

The impacts between this function and other functions are described in [4.2.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.2.3 Requirements

4.2.3.1 Licenses

There are no license requirements.

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

None

4.2.3.2.2 Mutually Exclusive Functions

RA T	Function Name	Function Switch	Reference
Lo w- freq uen cy TD D	Hyper Cell	NRDUCell.NrDu CellNetworking Mode set to HYPER_CELL	<i>Hyper Cell</i>
Lo w- freq uen cy TD D	Cell Combination	NRDUCell.NrDu CellNetworking Mode set to HYPER_CELL_CO MBINE_MODE , and NRDUCellMultiT rp.DataTransMo de set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>
Lo w- freq uen cy TD D	High-speed Railway Superior Experience	NRDUCell.HighS peedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>

4.2.3.2.3 Function Impacts

RA T	Function Name	Function Switch	Referenc e	Description
Lo w- freq uen cy TDD	Power saving BWP (low- frequency TDD)	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSwitch parameter	<i>UE Power Saving</i>	After power saving BWP is enabled, PDSCH scheduling is not fixed at the same start position and the position for PDSCH scheduling is partially staggered between indoor and outdoor cells. Consequently, the benefits provided by interference avoidance for indoor cells decrease. If power saving BWP causes the position for PDSCH scheduling to be completely staggered between indoor and outdoor cells, interference avoidance for indoor cells occasionally causes the average downlink throughput to decrease for these cells. Despite this, interference avoidance for indoor cells improves the downlink spectral efficiency of the entire network.
Lo w- freq uen cy TDD	Downlink interference randomization- based scheduling	DL_INTRF_RAN DOM_SWITCH option of the NRDUCellAlgorithm.AdaptiveEdgeExpEnhSwitch parameter	<i>Downlink Scheduling</i>	Only interference avoidance for indoor cells works for PDSCH scheduling when this function and downlink interference randomization-based scheduling are both enabled.

RA T	Function Name	Function Switch	Referenc e	Description
Lo w- freq uen cy TD D	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	When NRDUCellRes.DlMinRbStartPos is set to 0 , interference avoidance for indoor cells does not take effect.
Lo w- freq uen cy TD D	Enhanced interference avoidance for indoor cells	INDOOR_INTRF_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.IndoorOutdoorCooperateSw parameter	<i>3D Networking Experience Improvement</i>	Only enhanced interference avoidance for indoor cells takes effect when this function and interference avoidance for indoor cells are both enabled.

4.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.2.3.4 Others

- The Xn interface of the gNodeB is normal.
- Indoor cells and interfering neighboring cells cannot be deployed on the same base station.

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

Table 4-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Indoor Outdoor Cooperate Switch	NRDUCellAlgoSwitch.IndoorOutdoorCooperateSw	INDOOR_INTRF_AVOID_SW	Select this option for indoor cells that are affected by interference.

4.2.4.1.2 Using MML Commands

Before using MML commands, refer to **4.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling the function of interference avoidance for indoor cells  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, IndoorOutdoorCooperateSw = INDOOR_INTRF_AVOID_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the function of interference avoidance for indoor cells  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, IndoorOutdoorCooperateSw = INDOOR_INTRF_AVOID_SW-0;
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

Run the **LST NRDUCELLALGOSWITCH** command. If the following information is displayed in the command output, interference avoidance for indoor cells has been enabled.

Indoor Outdoor Cooperate Switch = Indoor Interference Avoidance Switch:On

After this function is enabled, use a drive test tool to observe the positions of indoor cells' PDSCH RBs. If the PDSCH RB allocation start position changes, interference avoidance for indoor cells has taken effect.

4.2.4.3 Network Monitoring

After this function is enabled, observe the following indicators to calculate the extent to which the downlink cell spectral efficiency and average downlink UE throughput have improved:

- Downlink spectral efficiency of a cell. This indicator can be calculated by using the following formula: $\frac{N.ThpVol.DL}{(N.PRBDL.DrbUsed.Avg \times N.DLPDSCH.Tti.Num)}$, expressed as bit/(TTI·PRB).
- **User Downlink Average Throughput (DU)**

NOTE

The downlink spectral efficiency of a cell can be expressed as bits per second per Hz by using the following formula: $\frac{N.ThpVol.DL}{(N.PRBDL.DrbUsed.Avg \times N.DLPDSCH.Tti.Num \times 2000/360000)}$. The idea behind this formula is as follows: 2000 is introduced for a TTI-to-second conversion, because one second has 2000 TTIs (1s = 1000 ms, 1 TTI = 0.5 ms). 360000 is introduced for a PRB-to-Hz conversion, because one PRB spans 360,000 Hz in the frequency domain (1 PRB = 12 subcarriers, 1 subcarrier = 30,000 Hz).

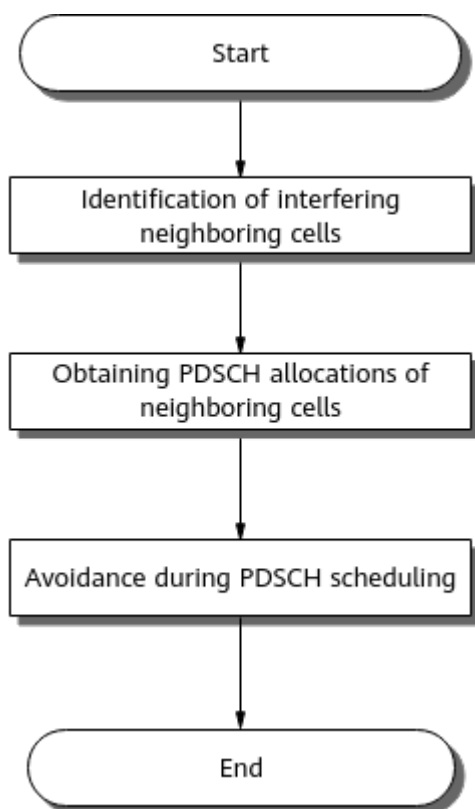
4.3 Enhanced Interference Avoidance for Indoor Cells

4.3.1 Principles

Interference occurs between an indoor cell and an outdoor cell when their RB positions for downlink scheduling overlap ([Figure 4-5](#)). Interference avoidance for indoor cells ([4.2 Interference Avoidance for Indoor Cells](#)) enables an indoor cell to adjust its PDSCH start position to shift its PDSCH away from the frequency-domain positions used by interfering neighboring cells, mitigating interference. However, this function requires the configuration of neighbor relationships—a process that may involve manual intervention. To simplify operation, enhanced interference avoidance for indoor cells is introduced, enabling automatic identification of interfering neighboring cells.

Enhanced interference avoidance for indoor cells is controlled by the **INDOOR_INTRF_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch.IndoorOutdoorCooperateSw** parameter. [Figure 4-8](#) shows how this function works.

Figure 4-8 Process of enhanced interference avoidance for indoor cells



1. Identification of interfering neighboring cells

The **INTRA_FREQ_NCELL_STAT_SW** option of the **NRDUCellCollabServ.NeighborNrCellMgmtSw** parameter specifies whether to enable intra-frequency neighboring cell measurement. After this option is selected, UEs report A3 measurements of intra-frequency neighboring cells based on the A3 threshold (specified by **NRCellComp.IntraFreqNCellStatA3Offset**). The indoor cell then ranks these neighboring cells by the measurements and selects the top five interfering neighboring cells. Note that this option needs to be selected for both the indoor and neighboring cells. Any neighboring cell with this option deselected is excluded from this ranking and subsequent procedures.

NOTE

- Interfering neighboring cells are identified every 30 minutes.
- After enhanced interference avoidance for indoor cells is enabled, the system first identifies interfering neighboring cells and then initiates interference avoidance once such cells are detected.

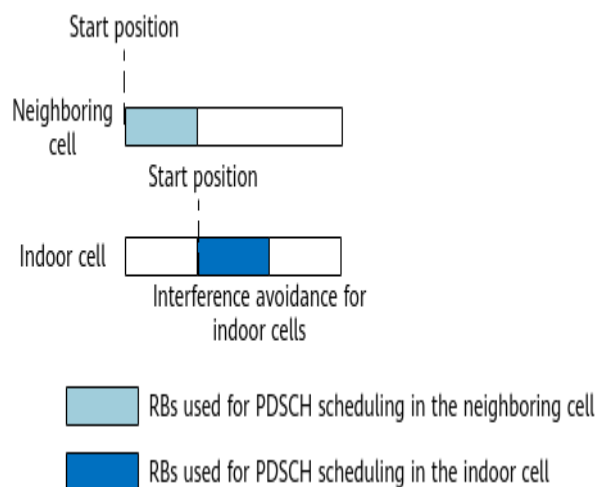
2. Obtaining PDSCH allocations of neighboring cells

The indoor cell obtains the PDSCH start-RB allocations of neighboring cells over the Xn interface.

3. Avoidance during PDSCH scheduling

The indoor cell schedules its PDSCH to start at a different RB position, shifting its PDSCH away from the PDSCH in those neighboring cells whenever possible, as shown in [Figure 4-9](#).

Figure 4-9 Avoidance during PDSCH scheduling



In this function, indoor cells are defined as cells served by LampSite base stations or macro base stations that provide DAS-based indoor coverage. (For details, see *Indoor Coverage*.) Note that indoor cells and interfering neighboring cells cannot be deployed on the same base station.

4.3.2 Network Analysis

4.3.2.1 Benefits

Deployment Suggestions

- It is recommended that this function be enabled in isolated indoor cells—each serving a single floor and typically experiencing interference from outdoor cells.
- If multiple indoor cells are deployed for contiguous coverage (with all neighboring cells being indoor cells), it is not recommended that this function be enabled, because interference between these cells predominates and cannot be effectively mitigated through this function.
- Additionally, it is not recommended that this function be enabled in outdoor cells.
- Given the same scenario, enhanced interference avoidance for indoor cells is preferred over interference avoidance for indoor cells (described in [4.2 Interference Avoidance for Indoor Cells](#)) because it automatically identifies interfering neighboring cells.

Benefits

This function mitigates interference between indoor and outdoor cells under light-load conditions with burst services if (1) downlink interference randomization-based scheduling is disabled or (2) downlink interference randomization-based scheduling is enabled and these cells have the same PCI mod 3 value. This function improves the indoor cell's downlink spectral efficiency and increases the average downlink throughput of UEs in the overlapping areas between these cells by 2%–3%.

NOTE

- Typically, a cell is considered lightly loaded if its downlink PRB usage is less than or equal to 30%. Downlink PRB usage of a cell = $(\text{N.PRB.DL.Used.Avg} / \text{N.PRB.DL.Avail.Avg}) \times 100\%$
- Downlink cell spectral efficiency = $\text{N.ThpVol.DL} / (\text{N.PRB.DL.DrbUsed.Avg} \times \text{N.DL.PDSCH.Tti.Num})$

To express the downlink spectral efficiency of a cell as bits per second per Hz, the preceding formula can be changed as follows: $\text{N.ThpVol.DL} / (\text{N.PRB.DL.DrbUsed.Avg} \times \text{N.DL.PDSCH.Tti.Num} \times 2000 / 360000)$. The idea behind this formula is as follows: 2000 is introduced for a TTI-to-second conversion, because one second has 2000 TTIs (1s = 1000 ms, 1 TTI = 0.5 ms). 360000 is introduced for a PRB-to-Hz conversion, because one PRB spans 360,000 Hz in the frequency domain (1 PRB = 12 subcarriers, 1 subcarrier = 30,000 Hz).

4.3.2.2 Impacts

The impacts between this function and other functions are described in [4.3.3.2.3 Function Impacts](#).

There is no impact on the network.

4.3.3 Requirements

4.3.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-060202	3D Networking Experience Improvement	NR0S03DNEM00	Per Cell
For the license control item NR0S03DNEM00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the INDOOR_INTRF_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.IndoorOutdoorCooperateSw parameter cannot be modified due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.			

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TD	Intra-frequency neighboring cell measurement	INTRA_FREQ_NCELL_STAT_SW option of the NRDUCellCollabServ.NeighborNrCellMgmtSw parameter	None	The switch for intra-frequency neighboring cell measurement must be turned on.

4.3.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
Low-frequency TD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>

RAT	Function Name	Function Switch	Reference
Low - frequency TD D	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE , and NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>
Low - frequency TD D	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>

4.3.3.2.3 Function Impacts

RA T	Function Name	Function Switch	Referenc e	Description
Lo w- freq uen cy TDD	Power saving BWP (low- frequency TDD)	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	After power saving BWP is enabled, PDSCH scheduling is not fixed at the same start position and the position for PDSCH scheduling is partially staggered between indoor and outdoor cells. Consequently, the benefits provided by enhanced interference avoidance for indoor cells decrease. If power saving BWP causes the position for PDSCH scheduling to be completely staggered between indoor and outdoor cells, enhanced interference avoidance for indoor cells occasionally causes the average downlink throughput to decrease for these cells. Despite this, enhanced interference avoidance for indoor cells improves the downlink spectral efficiency of the entire network.
Lo w- freq uen cy TDD	Downlink interference randomization- based scheduling	DL_INTRF_RAN DOM_SW option of the NRDUCellAlgoSwitch.Adaptive EdgeExpEnhSwitch parameter	<i>Downlink Scheduling</i>	Only enhanced interference avoidance for indoor cells works for PDSCH scheduling when this function and downlink interference randomization-based scheduling are both enabled.

RA T	Function Name	Function Switch	Referenc e	Description
Lo w- freq uen cy TD D	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	When NRDUCellRes.DlMinRbStartPos is set to 0 , enhanced interference avoidance for indoor cells does not take effect.
Lo w- freq uen cy TD D	Interference avoidance for indoor cells	INDOOR_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.IndoorOutdoorCooperateSw parameter	<i>3D Networking Experience Improvement</i>	Only enhanced interference avoidance for indoor cells takes effect when this function and interference avoidance for indoor cells are both enabled.

4.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

For NR TDD:

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.3.4 Cells

NR TDD: The serving cell and interfering neighboring cells must have the same bandwidth, slot configuration, CP, and SCS.

4.3.3.5 Others

- The Xn interface of the gNodeB is normal.
- Indoor cells and interfering neighboring cells cannot be deployed on the same base station.

4.3.4 Operation and Maintenance

4.3.4.1 Data Configuration

4.3.4.1.1 Data Preparation

Table 4-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Neighboring NR Cell Management Switch	NRDUCellCollabServ.NeighborNrCellMgmtSwitch	INTRA_FREQ_NCELL_STAT_SW	Select this option for indoor cells that are affected by interference.
Indoor Outdoor Cooperative Switch	NRDUCellAlgoSwitch.IndoorOutdoorCooperateSw	INDOOR_INTRF_AVOID_ENH_SW	Select this option for indoor cells that are affected by interference.
Intra-Freq Neighbor Cell Statistics A3 Offset	NRCellComp.IntraFreqNCellStatA3Offset	None	Retain the default value.

4.3.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling identification of interfering neighboring cells
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, NeighborNrCellMgmtSw=INTRA_FREQ_NCELL_STAT_SW-1;
//Enabling the function of enhanced interference avoidance for indoor cells
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, IndoorOutdoorCooperateSw =
INDOOR_INTRF_AVOID_ENH_SW-1;
//Configuring the IntraFreqNCellStatA3Offset parameter
MOD NRCELLCOMP: NrCellId=0, IntraFreqNCellStatA3Offset=-30;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the function of enhanced interference avoidance for indoor cells
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, IndoorOutdoorCooperateSw =
INDOOR_INTRF_AVOID_ENH_SW-0;
```

4.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.3.4.2 Activation Verification

Run the **LST NRDUCELLALGOSWITCH** command. If the following information is displayed in the command output, enhanced interference avoidance for indoor cells has been enabled.

```
Indoor Outdoor Cooperate Switch = Indoor Interference Avoidance Enh Sw:On
```

After this function is enabled, use a drive test tool to observe the positions of indoor cells' PDSCH RBs. If the PDSCH RB allocation start position changes, enhanced interference avoidance for indoor cells has taken effect.

4.3.4.3 Network Monitoring

After this function is enabled, monitor the following indicators to observe the extent to which the downlink cell spectral efficiency and average downlink UE throughput have improved:

- Downlink cell spectral efficiency = $\text{N.ThpVol.DL} / (\text{N.PRBDL.DrbUsed.Avg} \times \text{N.DLPDSCH.Tti.Num})$
- Average downlink UE throughput: **User Downlink Average Throughput (DU)**

 NOTE

To express the downlink spectral efficiency of a cell as bits per second per Hz, the preceding formula can be changed as follows: $\frac{N.ThpVol.DL}{(N.PRBDL.DrbUsed.Avg \times N.DLPDSCH.Tti.Num \times 2000/360000)}$. The idea behind this formula is as follows: 2000 is introduced for a TTI-to-second conversion, because one second has 2000 TTIs (1s = 1000 ms, 1 TTI = 0.5 ms). 360000 is introduced for a PRB-to-Hz conversion, because one PRB spans 360,000 Hz in the frequency domain (1 PRB = 12 subcarriers, 1 subcarrier = 30,000 Hz).

5 Inter-Frequency 3D Networking Experience Improvement

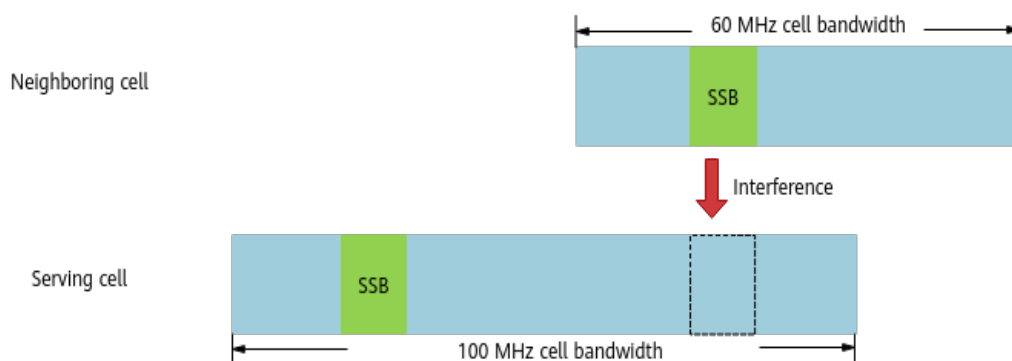
5.1 Principles

In indoor-outdoor or macro-pole continuous networking, if the serving cell and a neighboring cell work on different SSB frequencies, interference can occur in two situations, as described below:

- In the overlapping bandwidth, the serving cell's PDSCH experiences interference from the neighboring cell's SSB. [Figure 5-1](#) illustrates such interference in TDD.
- The serving cell's PDSCH experiences interference from the neighboring cell's CSI-RS when the two cells' CSI-RS transmissions are not time-aligned, as shown in [Figure 4-2](#). (A similar situation occurs when the two cells work on the same SSB frequency.)

To mitigate the preceding common-channel interference, inter-frequency 3D networking experience improvement is introduced. Based on the types of interfering signals, inter-frequency 3D networking experience improvement is classified into inter-frequency SSB interference avoidance, enhanced inter-frequency SSB interference avoidance, and inter-frequency CSI-RS interference avoidance.

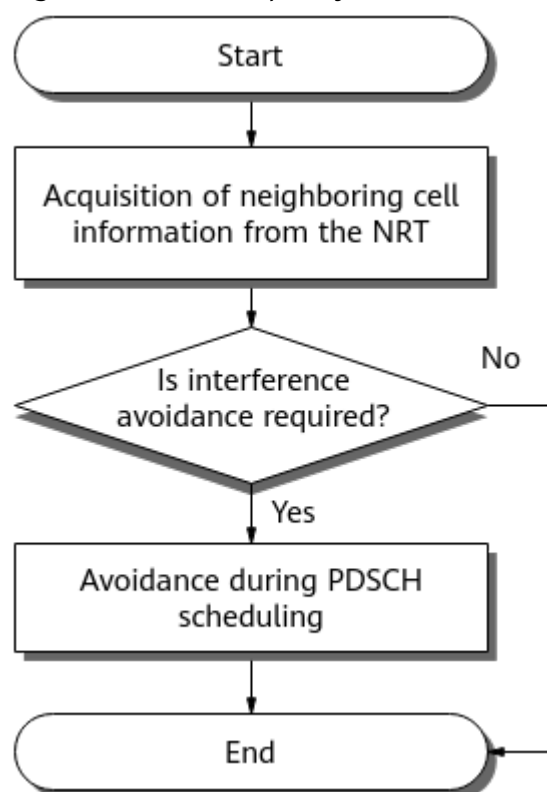
Figure 5-1 SSB interference from a neighboring cell in inter-frequency networking (TDD)



Inter-Frequency SSB Interference Avoidance

This function is controlled by the **INTER_FREQ_SSB_AVOID_SW** option of the **NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch** parameter. [Figure 5-2](#) shows how inter-frequency SSB interference avoidance works. With this function, the serving cell obtains information about neighboring cells from the neighboring relation table (NRT) and then their SSB configurations over the Xn interface. Based on this information, the gNodeB determines whether the serving cell is affected by common-channel interference from any of these neighboring cells. If such interference is detected, PDSCH scheduling for all UEs avoids the time-frequency positions used by that neighboring cell for SSB transmission.

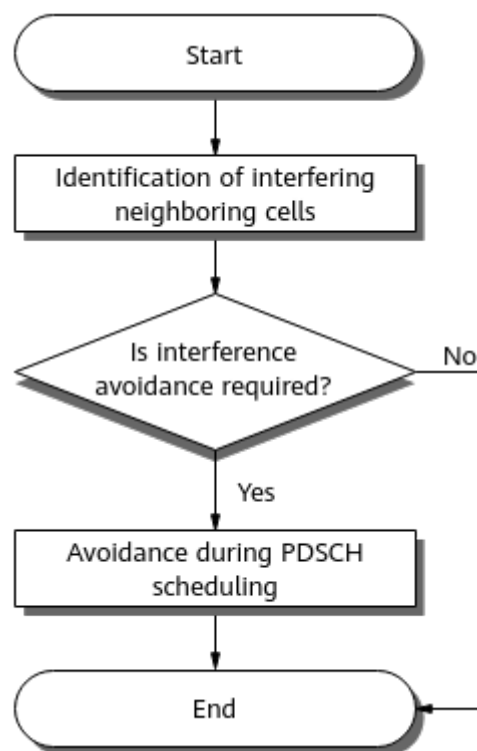
Figure 5-2 Inter-frequency SSB interference avoidance



Enhanced Inter-Frequency SSB Interference Avoidance

This function is controlled by the **INTER_FREQ_SSB_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch** parameter. [Figure 5-3](#) shows how enhanced inter-frequency SSB interference avoidance works.

Figure 5-3 Enhanced inter-frequency SSB interference avoidance



1. Identification of interfering neighboring cells

The **INTER_FREQ_NCELL_STAT_SW** option of the **NRDUCellCollabServ.NeighborNrCellMgmtSw** parameter specifies whether to enable inter-frequency neighboring cell measurement. After this option is selected, the serving cell uses UE-reported A3 or A5 measurements of inter-frequency neighboring cells to rank these cells, and then selects the top five interfering neighboring cells. Note that this option needs to be selected for both the serving and neighboring cells. Any neighboring cell with this option deselected is excluded from this ranking and subsequent procedures.

2. Determining whether interference avoidance is required

The serving cell obtains SSB configurations of a neighboring cell over the Xn interface. This occurs only if enhanced inter-frequency SSB interference avoidance is enabled for that neighboring cell and that neighboring cell is a common cell (not a hyper cell or combined cell). Based on the neighboring cell's SSB configurations, the serving cell determines whether its PDSCH experiences SSB interference from the neighboring cell as follows: If the SSB frequency-domain position of the neighboring cell is in the overlapping bandwidth, the serving cell's PDSCH experiences SSB interference from the neighboring cell, in which case interference avoidance is required. If the SSB frequency-domain position of the neighboring cell is not in the overlapping bandwidth, the serving cell's PDSCH does not experience SSB interference from the neighboring cell, in which case normal scheduling is applied.

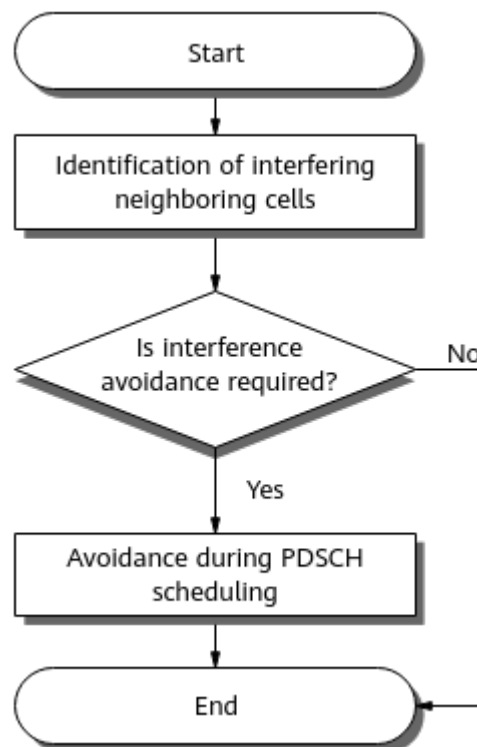
3. Avoidance during PDSCH scheduling

For full-buffer UEs, PDSCH scheduling avoids the time-frequency positions used by neighboring cells for SSB transmission whenever possible. For other UEs, interference avoidance is not required during scheduling.

Inter-Frequency CSI-RS Interference Avoidance

This function is controlled by the **INTER_FREQ_CSIRS_AVOID_SW** option of the **NRDUCellAlgoSwitch**. **CommChnIntrfAvoidSwitch** parameter. Figure 5-4 shows how inter-frequency CSI-RS interference avoidance works.

Figure 5-4 Inter-frequency CSI-RS interference avoidance



1. Identification of interfering neighboring cells

The **INTER_FREQ_NCELL_STAT_SW** option of the **NRDUCellCollabServ**. **NeighborNrCellMgmtSw** parameter specifies whether to enable inter-frequency neighboring cell measurement. After this option is selected, the serving cell uses UE-reported A3 or A5 measurements of inter-frequency neighboring cells to rank these cells, and then selects the top five interfering neighboring cells. Note that this option needs to be selected for both the serving and neighboring cells. Any neighboring cell with this option deselected is excluded from this ranking and subsequent procedures.

2. Determining whether interference avoidance is required

The serving cell obtains CSI-RS configurations of a neighboring cell over the Xn interface. This occurs only if inter-frequency CSI-RS interference avoidance is enabled for that neighboring cell and that neighboring cell is a common cell (not a hyper cell or combined cell). Based on the neighboring cell's CSI-RS configurations, the serving cell determines whether its PDSCH experiences CSI-RS interference from the neighboring cell as follows: If CSI-RS configurations are inconsistent between the two cells, the serving cell's PDSCH experiences CSI-RS interference from the neighboring cell, in which case interference avoidance is required. If CSI-RS configurations are consistent between the two cells, the serving cell's PDSCH does not experience CSI-RS interference from the neighboring cell, in which case normal scheduling is applied.

3. Avoidance during PDSCH scheduling

For all UEs, PDSCH scheduling avoids the time-frequency positions used by neighboring cells for CSI-RS transmission whenever possible.

 NOTE

- In inter-frequency CSI-RS interference avoidance and enhanced inter-frequency SSB interference avoidance, interfering neighboring cells are identified every 30 minutes.
- After inter-frequency CSI-RS interference avoidance or enhanced inter-frequency SSB interference avoidance is enabled, the system first identifies interfering neighboring cells and then initiates interference avoidance once such cells are detected.

5.2 Network Analysis

5.2.1 Benefits

Inter-Frequency SSB Interference Avoidance

In DTs in inter-frequency 3D continuous networking scenarios where bandwidths of adjacent cells overlap, you are advised to enable this function if all of the following conditions are met: The frequency-domain positions of SSBs are inconsistent between cells; the downlink BLERs of the serving cell in the slots with transmission of the neighboring cells' SSBs increase significantly (by 20%). If there is no SSB interference from neighboring cells, enabling this function may generate negative impacts. For details, see [5.2.2 Impacts](#).

- The downlink UE throughput increases when full buffer services are performed.
 - For stationary UEs whose PDSCHs suffer SSB interference from neighboring cells, the downlink throughput increases by 3% to 8%.
 - For UEs in DTs, the increase in the downlink throughput is related to the proportion of geographic locations where the UEs suffer SSB interference from neighboring cells in the DT routes. The larger the proportion, the higher the increase in the downlink throughput of UEs in DTs.
- When burst services are performed, the gains of this function are related to the time-frequency domain positions for scheduling. The specific gains cannot be determined, but there is no negative impact.

Enhanced Inter-Frequency SSB Interference Avoidance

Before enabling the inter-frequency SSB interference avoidance function, O&M personnel need to check for SSB interference from neighboring cells. In contrast, the enhanced inter-frequency SSB interference avoidance function automatically detects SSB interference from neighboring cells, making it a preferred function due to its lower O&M costs.

In DTs in inter-frequency 3D continuous networking scenarios where bandwidths of adjacent cells overlap, you are advised to enable enhanced inter-frequency SSB interference avoidance if all of the following conditions are met: The frequency-domain positions of SSBs are inconsistent between cells; the downlink BLERs of the serving cell in the slots with transmission of the neighboring cells' SSBs increase significantly (by 20%).

- The downlink UE throughput increases when full buffer services are performed.
 - For stationary UEs whose PDSCHs suffer SSB interference from neighboring cells, the average downlink throughput increases by 3% to 8%.
 - For UEs in DTs, the increase in the average downlink throughput is related to the proportion of geographic locations where the UEs suffer SSB interference from neighboring cells in the DT routes. The larger the proportion, the higher the increase in the average downlink throughput of UEs in DTs.
- When burst services are performed, no gains or negative impacts are generated because interference avoidance is not required during scheduling.

Inter-Frequency CSI-RS Interference Avoidance

It is recommended that inter-frequency CSI-RS interference avoidance be enabled in inter-frequency 3D continuous networking scenarios where bandwidths of adjacent cells overlap if all of the following conditions are met: The time-domain positions of cell-level periodic CSI-RSs are inconsistent between these cells; the downlink BLERs of the serving cell in the slots with transmission of the neighboring cells' CSI-RSs increase significantly (by 20%). This function reduces interference from neighboring cells and increases the downlink throughput of UEs in overlapping areas. The average downlink throughput of beneficiary UEs in the overlapping areas increases by up to 5%. The gains are related to such factors as the traffic model and network load. The larger the data packet size, the greater the gains. The higher the network load, the greater the gains. This function is recommended when the serving cell is in medium/heavy load (downlink PRB usage > 30%). The time-domain positions of cell-level periodic CSI-RSs are determined by **NRDUCellCsirs.CsiPeriod** and **NRDUCellCsirs.CsirsCellResourceNum**.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

- Inter-frequency SSB interference avoidance

Inter-frequency SSB interference avoidance reduces the time-frequency resources available for downlink scheduling due to the avoidance of SSB interference. The degree of the reduction depends on these configurations: cell bandwidth and neighboring cell's SSB resource quantity. For a TDD cell with a bandwidth of 60 MHz, available RBs are reduced by up to 14%. As such, the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) decreases by up to 14% for full buffer services.
- Enhanced inter-frequency SSB interference avoidance
 - A cell will automatically restart after the setting of the option for this function is changed. If this function is enabled for an indoor distributed base station or pole-mounted base station, it is recommended that this function also be enabled for adjacent macro base stations. Doing so lessens the impact of cell restarts on the network.
 - Enhanced inter-frequency SSB interference avoidance reduces the time-frequency resources available for downlink scheduling due to the

avoidance of SSB interference. The degree of the reduction depends on these configurations: cell bandwidth and neighboring cell's SSB resource quantity. For a TDD cell with a bandwidth of 60 MHz, available RBs are reduced by up to 14%. As such, the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) decreases by up to 14% for full buffer services.

- Inter-frequency CSI-RS interference avoidance
 - A cell will automatically restart after the setting of the option for this function is changed. If this function is enabled for an indoor distributed base station or pole-mounted base station, it is recommended that this function also be enabled for adjacent macro base stations. Doing so lessens the impact of cell restarts on the network.
 - Inter-frequency CSI-RS interference avoidance reduces the time-frequency resources available for downlink scheduling due to the avoidance of CSI-RS interference. The degree of the reduction depends on these configurations: numbers of CSI-RS ports for the serving cell and for neighboring cells, and difference in cell-level CSI-RS time-frequency resources between the serving and neighboring cells. Available RBs are reduced by up to 3%. As such, the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) decreases by up to 3% for full buffer services.

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-060202	3D Networking Experience Improvement	NR0S03DNEM00	Per Cell

Feature ID	Feature Name	Model	Sales Unit
For the license control item NR0S03DNEM00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the INTER_FREQ_SSB_AVOID_SW, INTER_FREQ_SSB_AVOID_ENH_SW, or INTER_FREQ_CSIRS_AVOID_SW option of the NRDUCellAlgoSwitch.<i>CommChnIntrfAvoidSwitch</i> parameter cannot be modified due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.<i>LicenseChkSw</i> parameter.			

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

Inter-Frequency SSB Interference Avoidance

None

Enhanced Inter-Frequency SSB Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	Inter-frequency neighboring cell measurement	INTER_FREQ_N CELL_STAT_SW option of the NRDUCellColla bServ.Neighbo rNrCellMgmtS w parameter	None	The switch for inter- frequency neighboring cell measurement must be turned on.
Lo w- freq uen cy TD D	Coverage- based inter- frequency handover	COVERAGE_BA SED_HO option of the NRCCellAlgoSwi tch.InterFreqH oSwitch parameter	<i>SA Mobility Managem ent in Connected Mode or NSA Mobility Managem ent</i>	The coverage-based inter-frequency handover function must be enabled.

Inter-Frequency CSI-RS Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	Inter-frequency neighboring cell measurement	INTER_FREQ_N CELL_STAT_SW option of the NRDUCellColla bServ.Neighbo rNrCellMgmtS w parameter	None	The switch for inter- frequency neighboring cell measurement must be turned on.
Lo w- freq uen cy TD D	Coverage- based inter- frequency handover	COVERAGE_BA SED_HO option of the NRCCellAlgoSwi tch.InterFreqH oSwitch parameter	<i>SA Mobility Managem ent in Connected Mode or NSA Mobility Managem ent</i>	The coverage-based inter-frequency handover function must be enabled.

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	CSI-RS rate matching	CSIRS_RATEM ATCH_SW option of the NRDUCellPdsc h.RateMatchS witch parameter	<i>Downlink Scheduling</i>	CSI-RS rate matching must be enabled.

5.3.2.2 Mutually Exclusive Functions

Inter-Frequency SSB Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	DL Blanking	NRDUCell.Peer CellSlotAssign ment	<i>Standards Complianc e</i>	The NRDUCell.PeerCellSlot Assignment parameter must be set to NOT_CONFIG .

Enhanced Inter-Frequency SSB Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	DL Blanking	NRDUCell.Peer CellSlotAssign ment	<i>Standards Complianc e</i>	The NRDUCell.PeerCellSlot Assignment parameter must be set to NOT_CONFIG .
Lo w- freq uen cy TD D	Inter-TRP SSB staggering	INTER_TRP_SS B_STAGGER_S WITCH option of the NRDUCellMult iTrp.MultiTrpA lgoSwitch parameter	<i>Hyper Cell Cell Combinati on</i>	These two functions are mutually exclusive: inter-TRP SSB staggering and enhanced inter- frequency SSB interference avoidance.

Inter-Frequency CSI-RS Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	DL Blanking	NRDUCell.Peer CellSlotAssign ment	<i>Standards Complianc e</i>	The NRDUCell.PeerCellSlot Assignment parameter must be set to NOT_CONFIG .
Lo w- freq uen cy TD D	Time-domain reservation mode	NRDUCellTime Intrf.TimeDom ainRsvMode	None	Inter-frequency CSI-RS interference avoidance cannot be enabled when the NRDUCellTimeIntrf.Ti meDomainRsvMode parameter is set to TIME_DOMAIN_RESERV ED_MODE_1 , TIME_DOMAIN_RESERV ED_MODE_2 , or TIME_DOMAIN_RESERV ED_MODE_3 .

5.3.2.3 Function Impacts

Inter-Frequency SSB Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	Enhanced inter- frequency SSB interference avoidance	INTER_FREQ_S SB_AVOID_EN H_SW option of the NRDUCellAlgo Switch.CommC hnIntrfAvoidS witch parameter	<i>3D Networkin g Experien ce Improvem ent</i>	Only enhanced inter- frequency SSB interference avoidance takes effect when this function and inter- frequency SSB interference avoidance are both enabled.

Enhanced Inter-Frequency SSB Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	Downlink intra- band CA	INTRA_BAND_ CA_SW option of the NRDUCellAlgo Switch.CaAlgo Switch parameter	<i>Carrier Aggregatio n</i>	When downlink intra- band CA is enabled, enhanced inter- frequency SSB interference avoidance does not take effect for CA UEs in their SCells.
Lo w- freq uen cy TD D	Downlink intra- FR inter-band CA	INTRA_FR_INT ER_BAND_CA_ SW option of the NRDUCellAlgo Switch.CaAlgo Switch parameter	<i>Carrier Aggregatio n</i>	When downlink intra-FR inter-band CA is enabled, enhanced inter-frequency SSB interference avoidance does not take effect for CA UEs in their SCells.
Lo w- freq uen cy TD D	Inter-gNodeB CA	INTER_GNODE B_CA_SW option of the NRDUCellAlgo Switch.CaAlgo Switch parameter	<i>Carrier Aggregatio n</i>	When inter-gNodeB CA is enabled, enhanced inter-frequency SSB interference avoidance does not take effect for CA UEs in their SCells.
Lo w- freq uen cy TD D	Inter-frequency SSB interference avoidance	INTER_FREQ_S SB_AVOID_SW option of the NRDUCellAlgo Switch.CommC hnIntrfAvoidS witch parameter	<i>3D Networkin g Experience Improvement</i>	Only enhanced inter- frequency SSB interference avoidance takes effect when this function and inter- frequency SSB interference avoidance are both enabled.
Lo w- freq uen cy TD D	Cell Combination	NRDUCell.NrD uCellNetworki ngMode set to HYPER_CELL_C OMBINE_MOD E	<i>Cell Combinati on</i>	Enhanced inter- frequency SSB interference avoidance does not take effect in combined cells.

Inter-Frequency CSI-RS Interference Avoidance

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	Downlink intra- band CA	INTRA_BAND_ CA_SW option of the NRDUCellAlgo Switch.CaAlgo Switch parameter	<i>Carrier Aggregatio n</i>	When downlink intra- band CA is enabled, inter-frequency CSI-RS interference avoidance does not take effect for CA UEs in their SCells.
Lo w- freq uen cy TD D	Downlink intra- FR inter-band CA	INTRA_FR_INT ER_BAND_CA_ SW option of the NRDUCellAlgo Switch.CaAlgo Switch parameter	<i>Carrier Aggregatio n</i>	When downlink intra-FR inter-band CA is enabled, inter-frequency CSI-RS interference avoidance does not take effect for CA UEs in their SCells.
Lo w- freq uen cy TD D	Inter-gNodeB CA	INTER_GNODE B_CA_SW option of the NRDUCellAlgo Switch.CaAlgo Switch parameter	<i>Carrier Aggregatio n</i>	When inter-gNodeB CA is enabled, inter- frequency CSI-RS interference avoidance does not take effect for CA UEs in their SCells.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

For NR TDD:

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Cells

The bandwidths of the serving and interfering neighboring cells must overlap.

Inter-frequency CSI-RS interference avoidance requires that the **NRDUCell.SlotAssignment** parameter not be set to **3_7_DSUUUUSUUU**.

5.3.5 Others

- The Xn interface of the gNodeB is normal.
- Inter-frequency SSB interference avoidance requires that the versions of all inter-frequency neighboring cells in the NRT of the serving cell be V100R018C00SPC010 or later.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Neighboring NR Cell Management Switch	NRDUCellCollabServ.NeighborNrCellMgmtSw	INTER_FREQ_NCELL_STAT_SW	Select this option for both the serving cell and neighboring cells.
Common Channel Interference Avoid Switch	NRDUCellAlgoSwitch.h.CommChnIntrfAvoidSwitch	INTER_FREQ_SSB_AVOID_SW	To enable inter-frequency SSB interference avoidance, select this option for the serving cell.
Common Channel Interference Avoid Switch	NRDUCellAlgoSwitch.h.CommChnIntrfAvoidSwitch	INTER_FREQ_SSB_AVOID_ENH_SW	To enable enhanced inter-frequency SSB interference avoidance, select this option for both the serving cell and neighboring cells.

Parameter Name	Parameter ID	Option	Setting Notes
Common Channel Interference Avoid Switch	NRDUCellAlgoSwitch.h.CommChnIntrfAvoidSwitch	INTER_FREQ_CSIRS_AVOID_SW	To enable inter-frequency CSI-RS interference avoidance, select this option for both the serving cell and neighboring cells.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling inter-frequency neighboring cell measurement
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, NeighborNrCellMgmtSw=INTER_FREQ_NCELL_STAT_SW-1;

//Enabling inter-frequency SSB interference avoidance
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CommChnIntrfAvoidSwitch=INTER_FREQ_SSB_AVOID_SW-1;

//Enabling enhanced inter-frequency SSB interference avoidance (Changing the setting of the
INTER_FREQ_SSB_AVOID_ENH_SW option will cause the cell to automatically restart.)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CommChnIntrfAvoidSwitch=INTER_FREQ_SSB_AVOID_ENH_SW-1;

//Enabling inter-frequency CSI-RS interference avoidance (Changing the setting of the
INTER_FREQ_CSIRS_AVOID_SW option will cause the cell to automatically restart.)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CommChnIntrfAvoidSwitch=INTER_FREQ_CSIRS_AVOID_SW-1;
```

NOTE

In inter-frequency neighboring cell measurement, enhanced inter-frequency SSB interference avoidance, and inter-frequency CSI-RS interference avoidance, the relevant options need to be selected for both the serving and neighboring cells.

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling inter-frequency SSB interference avoidance
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CommChnIntrfAvoidSwitch=INTER_FREQ_SSB_AVOID_SW-0;

//Disabling enhanced inter-frequency SSB interference avoidance (Changing the setting of the
INTER_FREQ_SSB_AVOID_ENH_SW option will cause the cell to automatically restart.)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
```

5.4.1.3 Using the MAE-Deployment

 NOTE

5.4.2 Activation Verification

-  **NOTE**

- After inter-frequency CSI-RS interference avoidance is enabled, start Uu signaling tracing as follows: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**. Check the RRCReconfiguration message tracing result. If the values of those (shown in the red box in [Figure 5-5](#)) in the *CSI-RS-ResourceMapping* IE of the RRC signaling ZP-CSI-RS-Resource change, this function has taken effect.

```

CSI-RS-ResourceMapping ::= SEQUENCE {
    frequencyDomainAllocation CHOICE {
        row1 BIT STRING (SIZE (4)),..
        row2 BIT STRING (SIZE (12)),..
        row4 BIT STRING (SIZE (3)),..
        other BIT STRING (SIZE (6))..
    },..
    nrofPorts ENUMERATED {p1,p2,p4,p8,p12,p16,p24,p32},..
    firstOFDMSymbolInTimeDomain INTEGER (0..13),..
    firstOFDMSymbolInTimeDomain2 INTEGER (2..12)
    -- OPTIONAL, -- Need R
    cdm-Type ENUMERATED {noCDM, fd-CDM2, cdm4-FD2-TD2, cdm8-FD2-TD4},..
    density CHOICE {
        dot5 ENUMERATED {evenPRBs, oddPRBs},..
        one NULL, ..
        three NULL, ..
        spare NULL..
    },..
    freqBand CSI-FrequencyOccupation, ..
    ...
}

```

5.4.3 Network Monitoring

54

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Carrier Aggregation*
- *Interference Avoidance*
- *Cell Combination*
- *High Speed Mobility*
- *Standards Compliance*
- *Hyper Cell*
- *Downlink Scheduling*
- *SA Mobility Management in Connected Mode*
- *NSA Mobility Management*
- 3GPP TS 38.331: "Radio Resource Control (RRC) protocol specification"
- *Beam Management*
- *Channel Management*
- *UE Power Saving*
- *Indoor Coverage*
- *URLLC*
- *Energy Conservation and Emission Reduction*
- *License Control Item Lists*